Introduction to computer vision

What is computer vision?

https://www.shutterstock.com/image-photo/berlin-germany-aug-30-2015-flea-340880915

What objects do you see in the scene?

What type of event is this?


Do you know which people are friends?

Where is the person taking the photo standing?


What does a computer see?

0	3	2	5	4	7	6	9	8	0	3	2	5	4	7	6	9	8	0	3	2	5	4	7	6	9	8	0	3	2	5	4	7	6	9	8	0	3	2	5	4	7	6	9	8
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What a computer sees

So how are we able to to draw out so much information from just one image?

Can we get a computer to understand this image in the same way that we do?

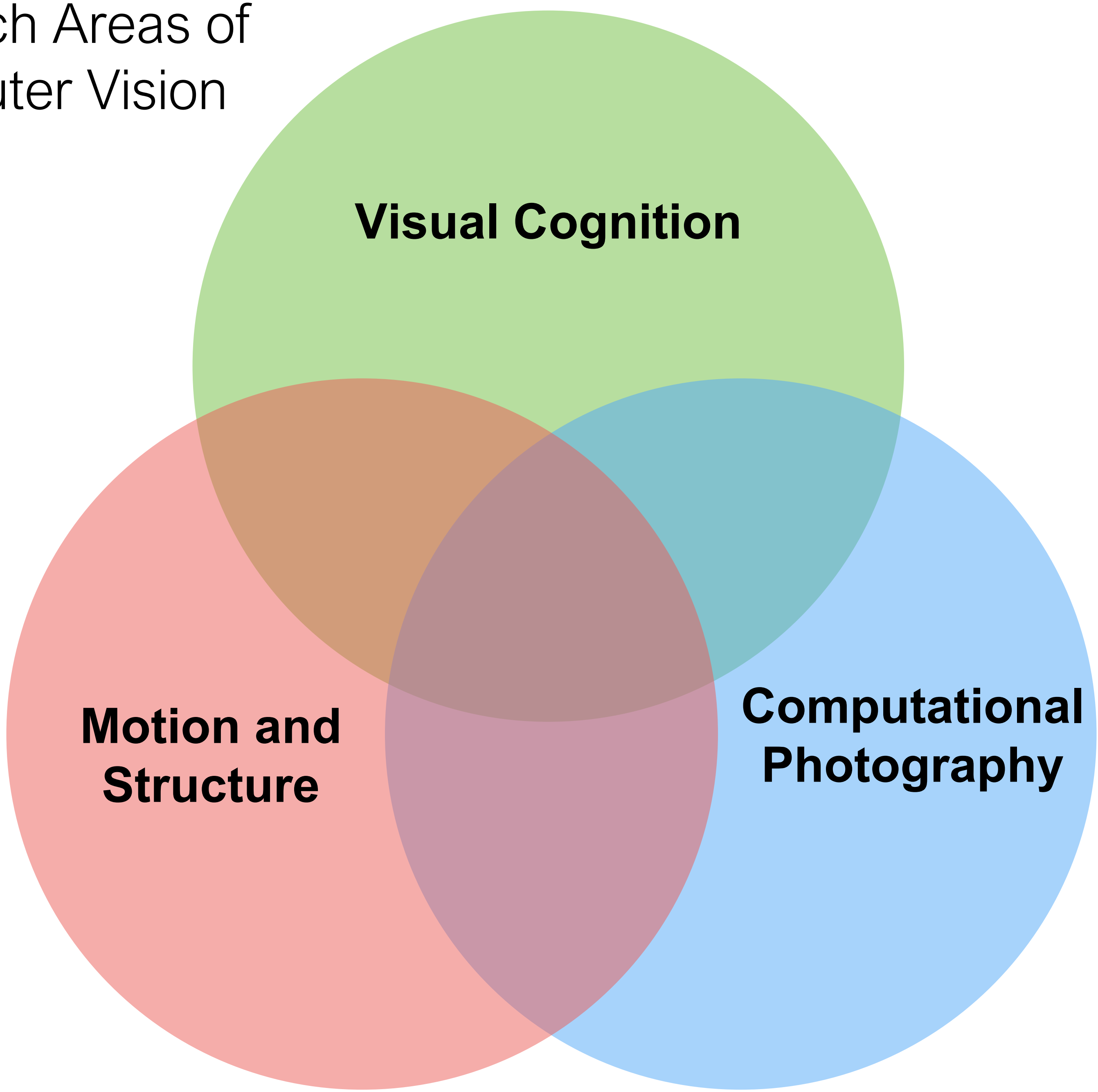
The goal of computer vision is

to give computers

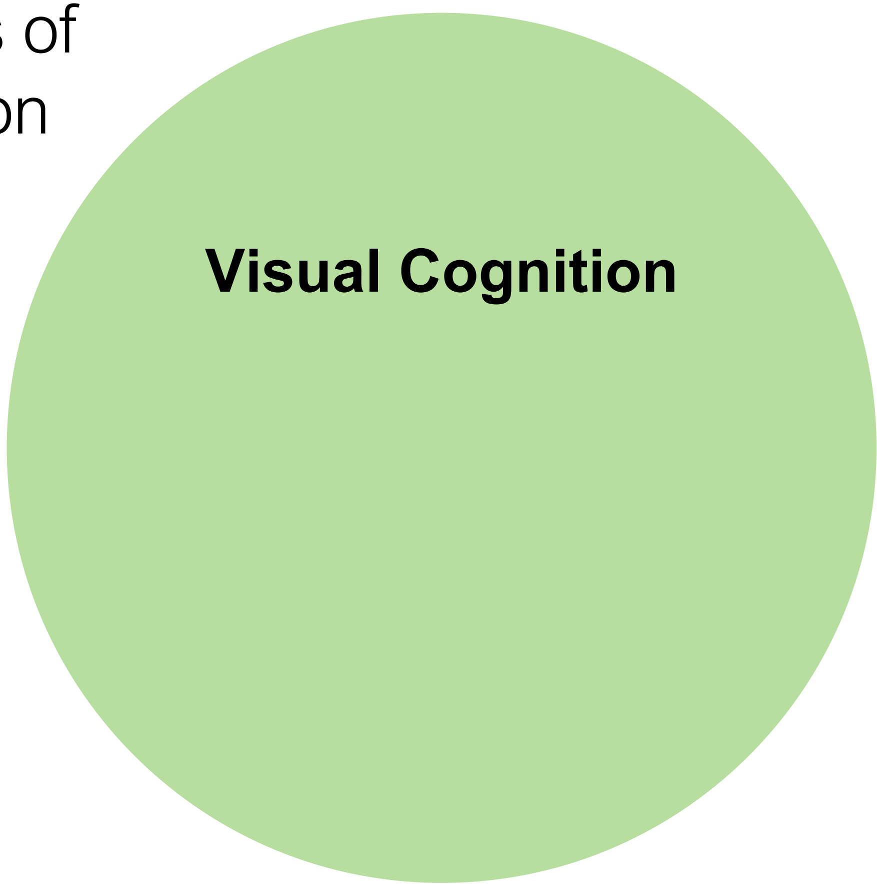
(super) human-level visual perception

Overview of Research Areas in Computer Vision

Research Areas of Computer Vision



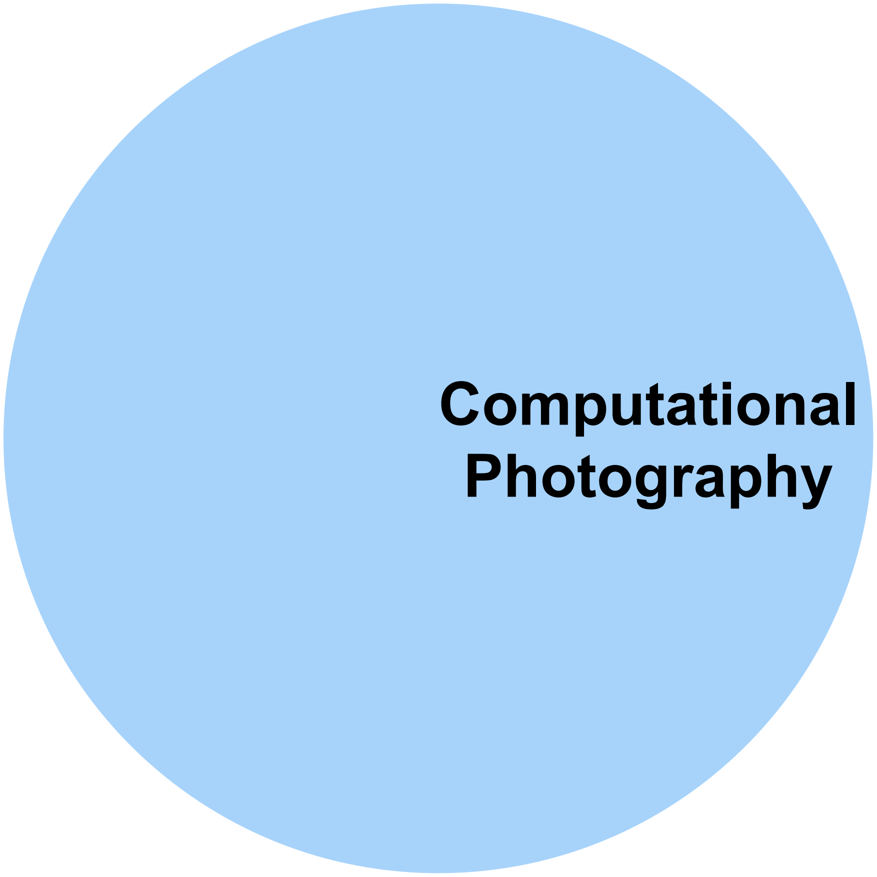
Research Areas of Computer Vision



What is in an image?

Research Areas of
Computer Vision

*How is an image
created?*



Research Areas of
Computer Vision

Motion and Structure

How does an image change as a result of motion?

Visual Cognition

What is in an image?

Segmentation and detection

Human detection

Object detection

Semantic segmentation

Super-pixels

Activity detection

Recognition

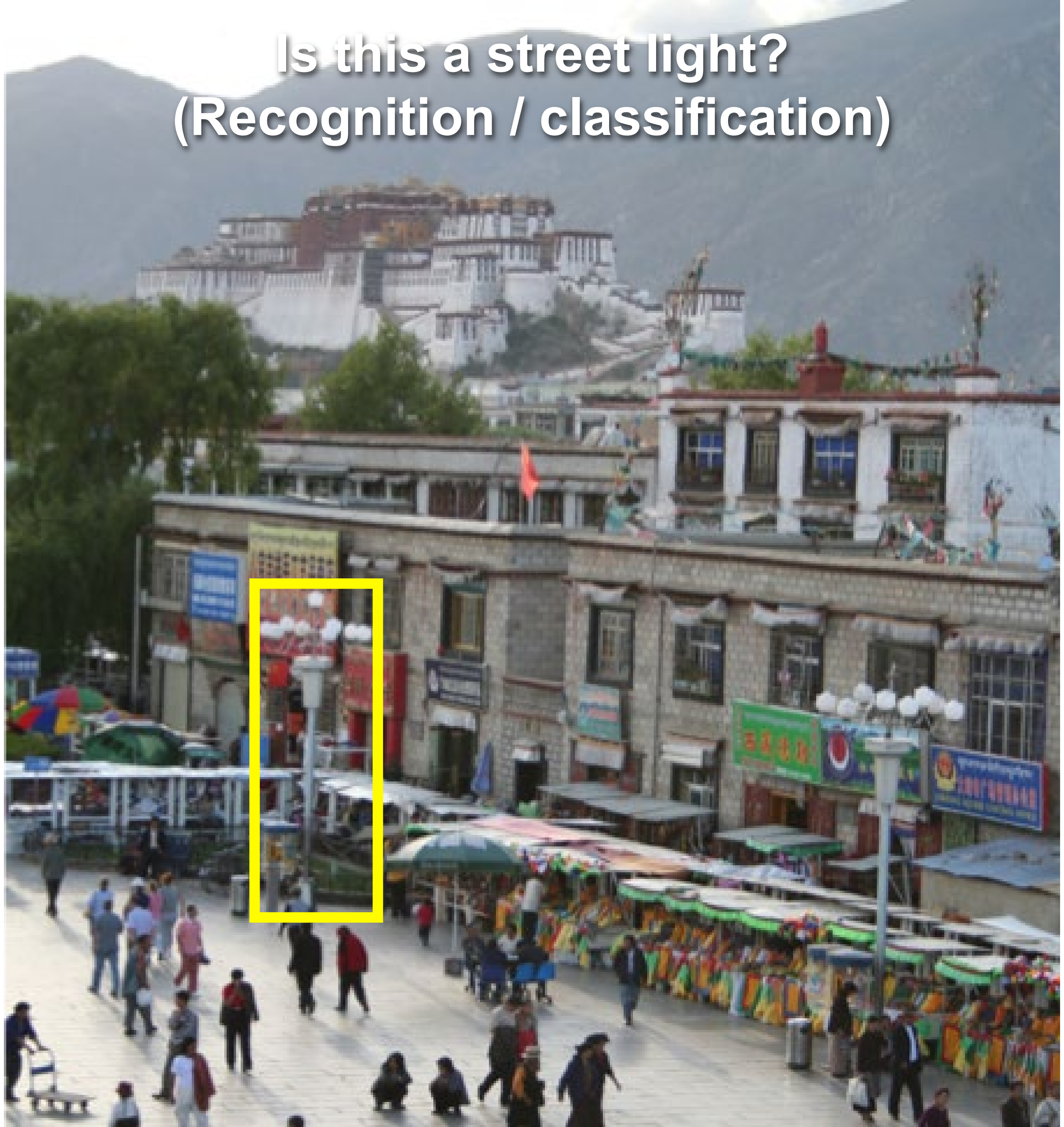
Human identification

Face recognition

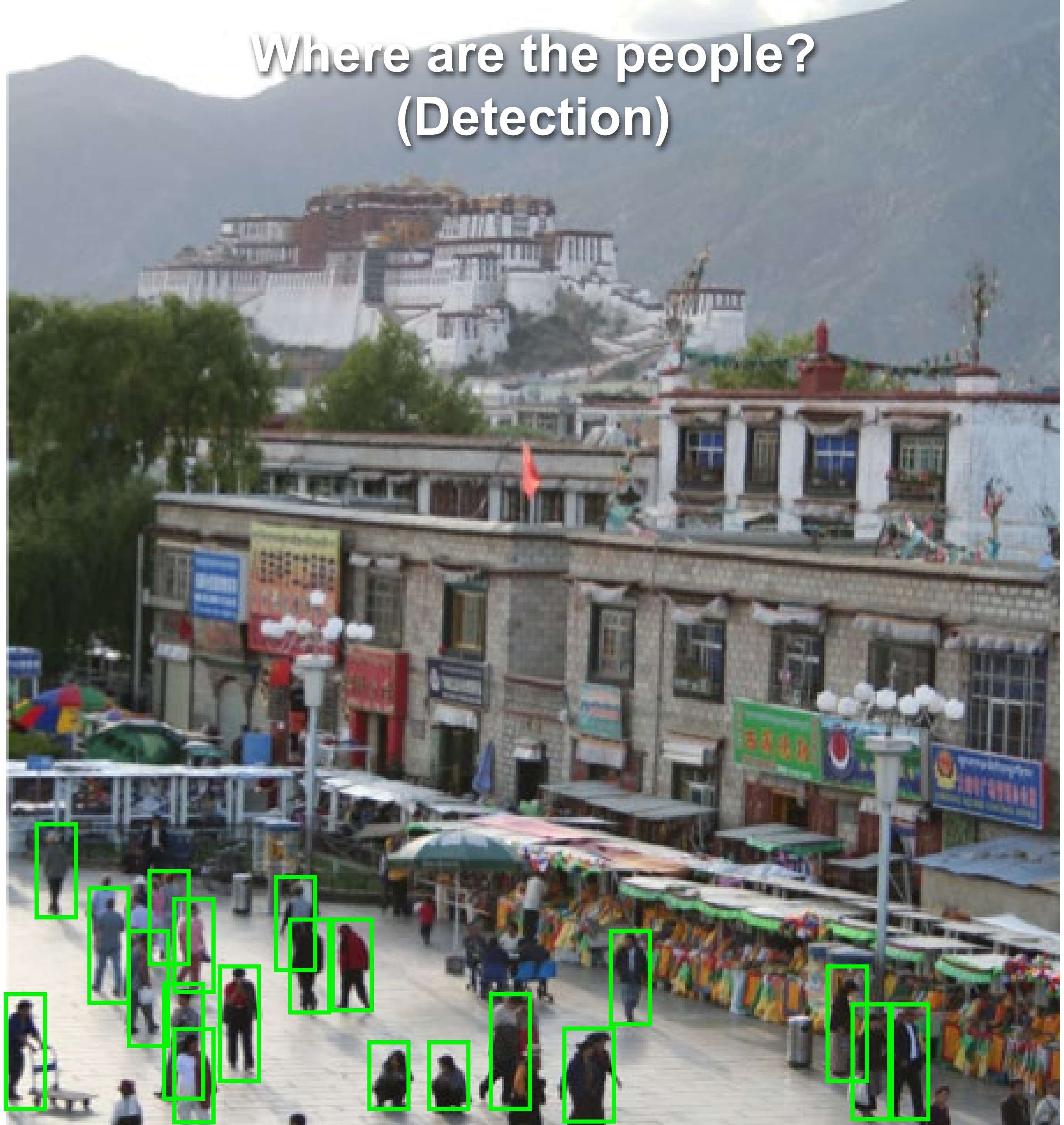
Object recognition

Activity recognition

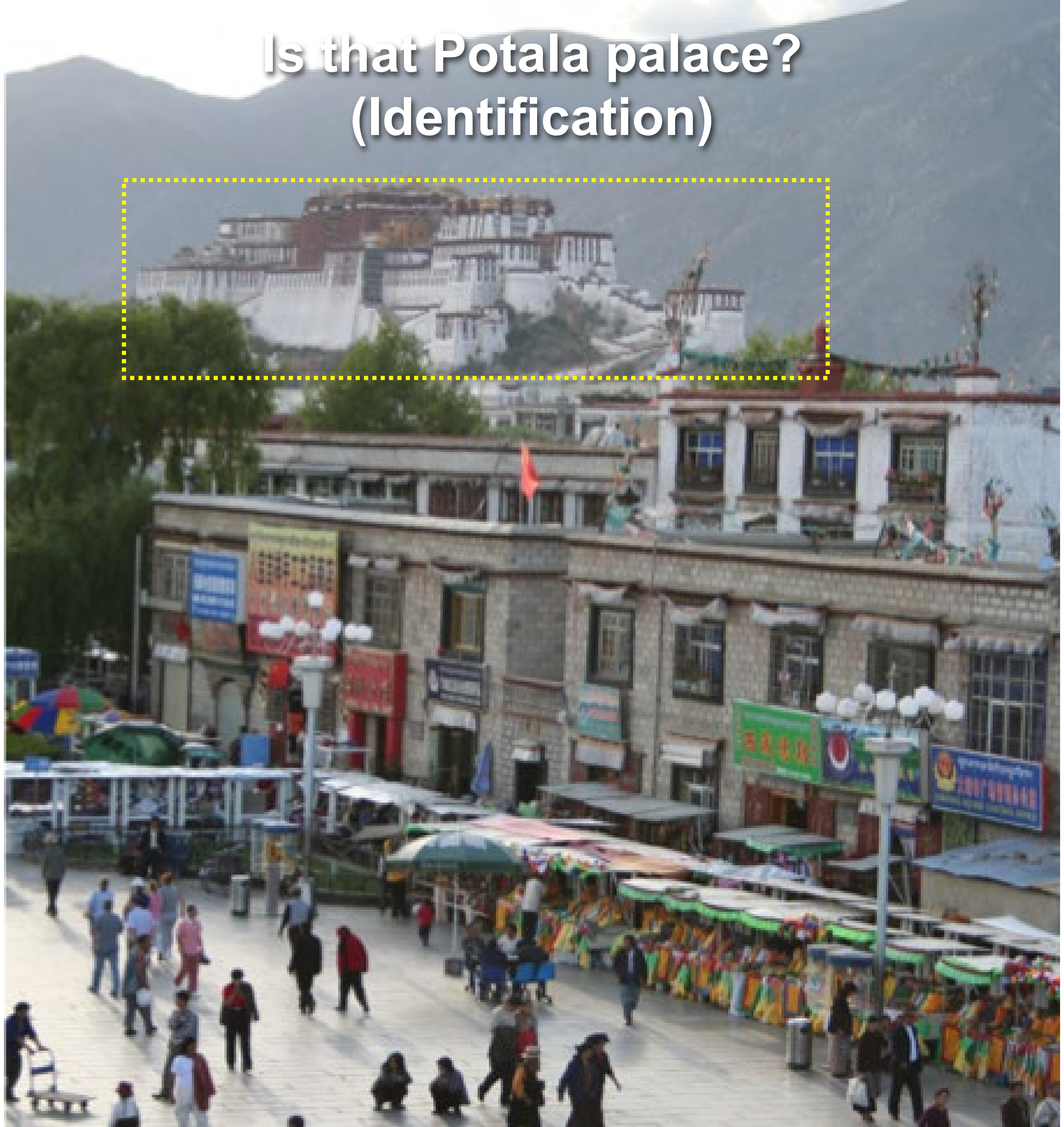
Is this a street light?
(Recognition / classification)

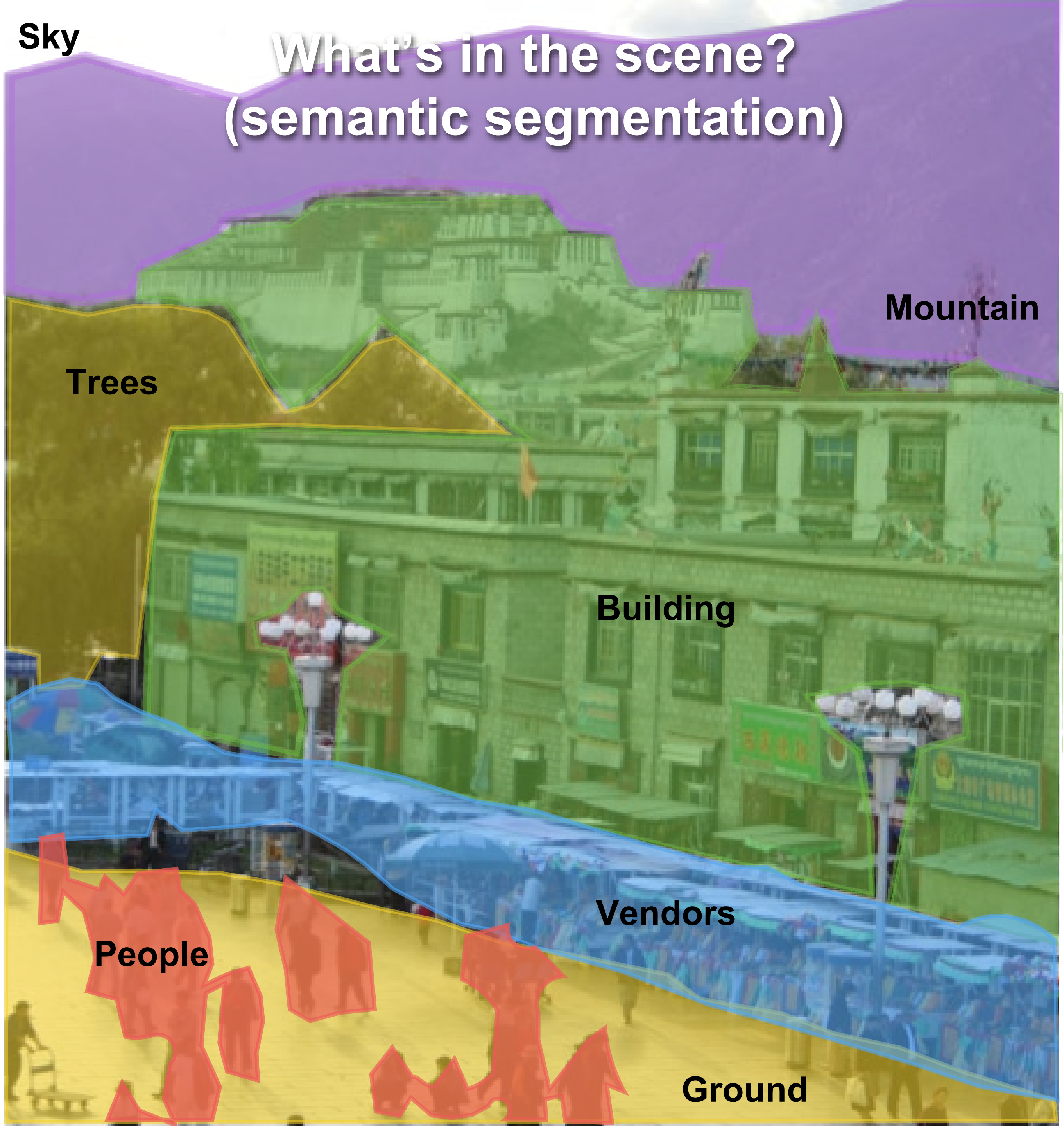


Where are the people?
(Detection)



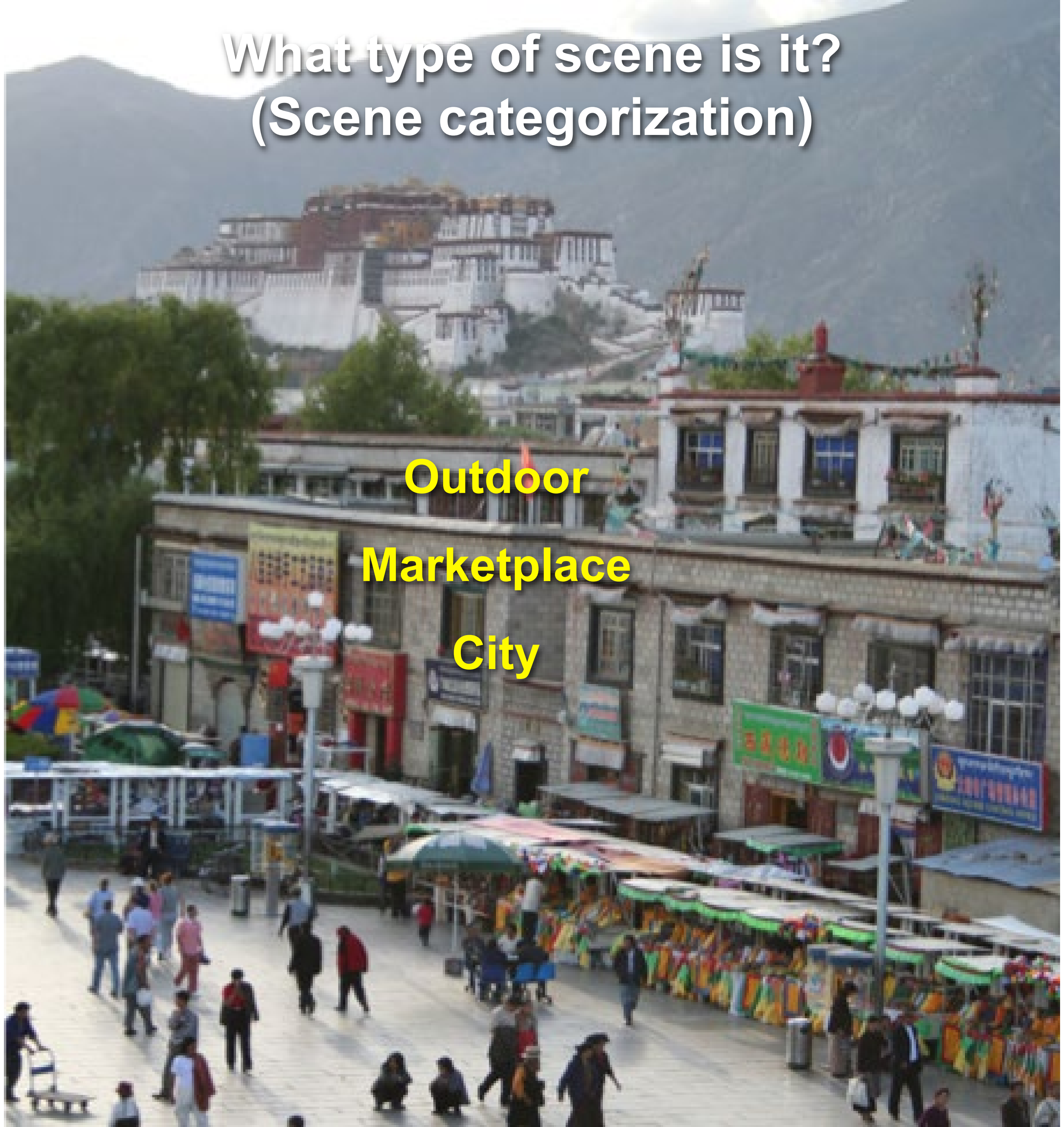
Is that Potala palace?
(Identification)



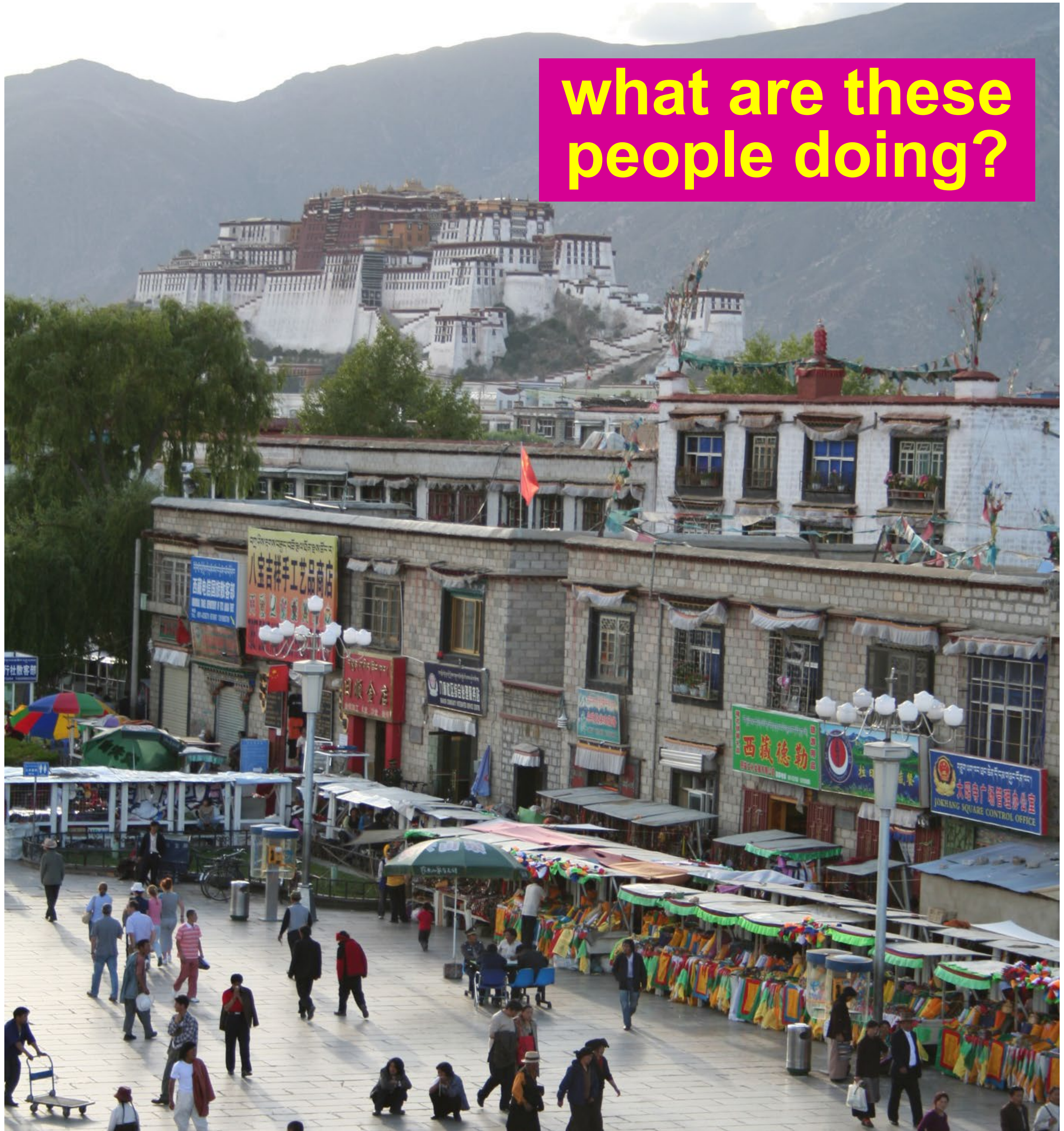

Sky
What's in the scene?
(semantic segmentation)
Mountain
Trees
Building
Vendors
People
Ground

Object categorization

What type of scene is it?
(Scene categorization)



Activity / Event Recognition



Computational Photography

How is an image created?

Computational Imaging

Light Field Cameras

Non-Line of Sight Imaging

Structured Light Cameras

Time of Flight Imaging

Computational Illumination and Display

Image and Video Processing

Computational Optics

Imaging Models

Event-based Cameras

High Dynamic Range Imaging

Synthesizing Image

In-painting

Denoising

Image Morphing 变形

Image Restoration

Image Matting and Segmentation 修边

Super-resolution and digital zoom

Colorization

Style Transfer

Image Generation

Image-based Rendering

Physics-based Rendering

Neural Rendering

Synthetic Defocus

Deblurring

Image Fusion

Synthesizing Images

In-painting

Denoising

Image Morphing

Image Restoration

Image Matting

Deblurring

Super-resolution

Colorization

Style Transfer

Image Generation

Image-based Rendering

Physics-based Rendering

Neural Rendering

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Computational Imaging

Light Field Cameras

Non-Line of Sight Imaging

Structured Light Cameras

Time of Flight Imaging

Computational Illumination and Display

Image and Video Processing

Computational Optics

Imaging Models

Event-based Cameras

High Dynamic Range Imaging


shutterstock

Motion and Structure

How does an image change when there is motion?

Multiple-view Geometry

Pose Estimation

Triangulation

Depth Estimation

Stereo correspondences

Camera calibration

3D reconstruction

Simultaneous Localization and Mapping (SLAM)

Visual Odometry 里程计

Monocular Depth Estimation

Surface normal estimation

State Estimation

Pedestrian Tracking

Face Tracking

Hand Tracking

Multi-object Tracking

Registration

Optical Flow

Image Alignment

Active Appearance Models

Image Registration

Non-rigid Motion Estimation

Motion Modeling

Human Pose and Motion Estimation

3D hand Pose Estimation

Trajectory analysis

Multi-agent trajectory forecasting

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shutterstock


Exchange Square
Next Step

Multiple-view Geometry

Pose Estimation

Triangulation

Depth Estimation

Stereo correspondences

Camera calibration

3D reconstruction

Simultaneous Localization and Mapping (SLAM)

Visual Odometry

Monocular Depth Estimation

State Estimation

Pedestrian Tracking

Face Tracking

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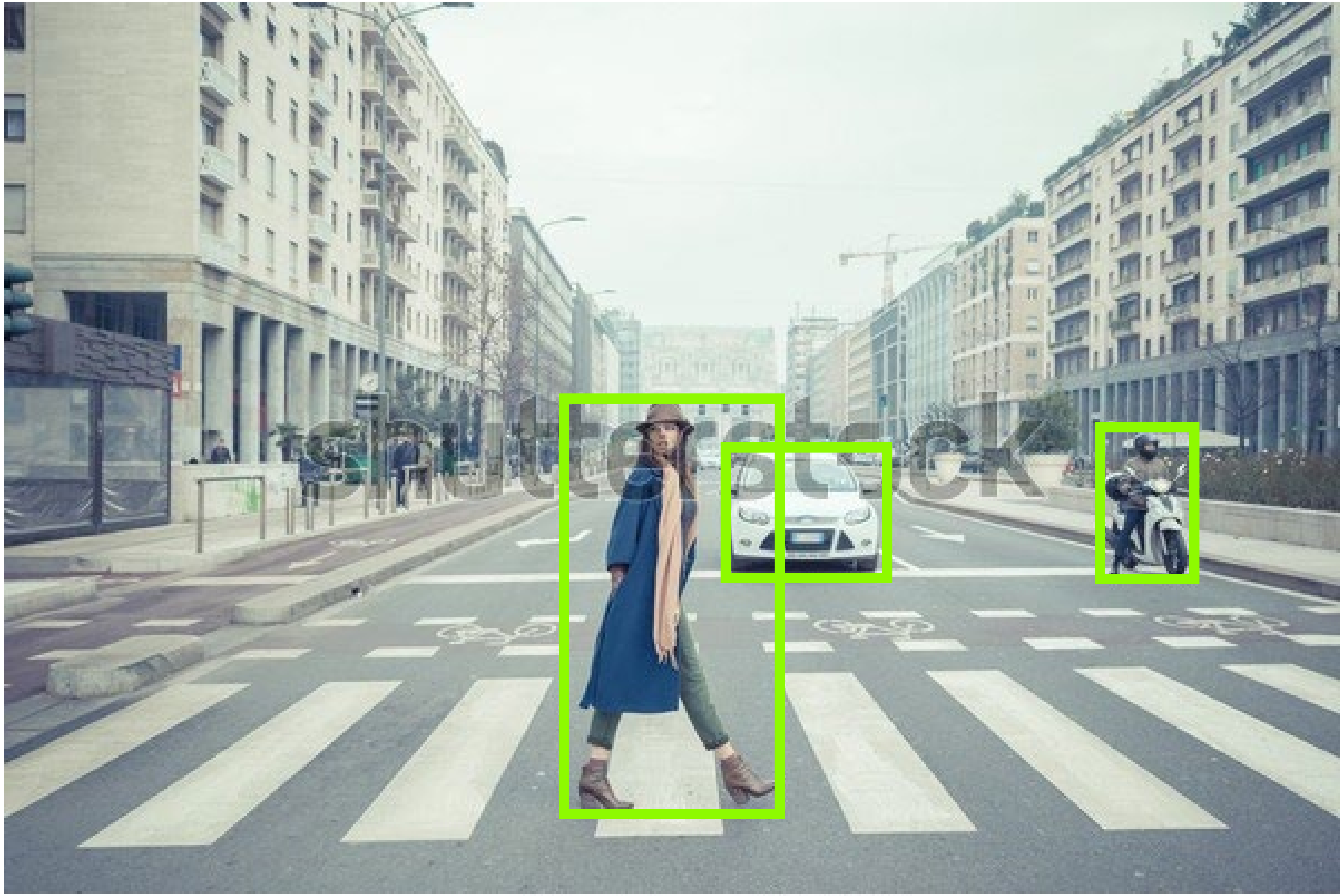
Trajectory analysis

Multi-agent trajectory forecasting



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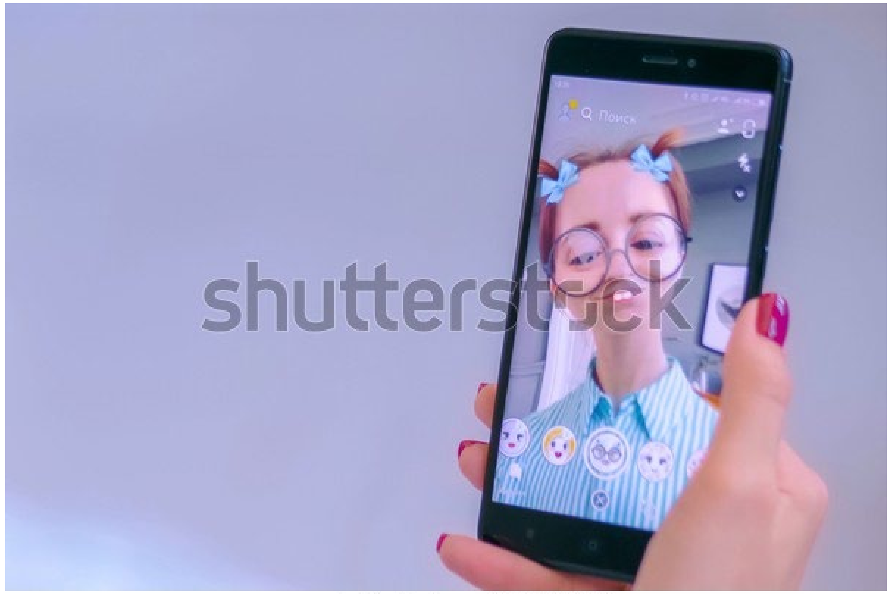

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Research Areas of
Computer Vision